

Paper Title

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Abstract—

- **Problem:** improve settling prediction for a position-dependent dual gantry without surrendering the meaning of its physical parameters
- **Method:** combine a quasi-LPV first-principles baseline, dynamic parallel LFR augmentation, closed-loop rollout training, and state-level orthogonal projection
- **Evidence:** compare baseline, unconstrained augmentation, orthogonal variants, and a black-box model on the completed validation ladder, including relevant motion and settling where available
- **Main result:** report predictive improvement together with the measured limits from short-horizon training, inactive added states, and non-orthogonal true missing dynamics
- **Conclusion:** state when the proposed decomposition is useful and which conditions prevent interpreting it as true parameter recovery

[TODO: Write the numerical abstract only after the final simulation and Telica result tables are frozen.]

Index Terms—grey-box identification, LPV systems, model augmentation, orthogonal projection, dual gantry

I. INTRODUCTION

[TODO: Kijken bronnen origineel van roland + hoekstra + bendeguz + roel drenth. currently i dont really state anything about orthogonality for interpretability for example.]

A. Industrial motivation and modelling gap

- Settling prediction limits throughput in coupled high-precision gantries, so model error has a direct operational cost [1]
- The concrete digital-twin use case is to evaluate relevant ASMPT motion and a changed controller before deployment on the machine
- The physics model exposes masses, stiffnesses, damping, and coupling, but omits position-dependent flexible dynamics that matter for prediction
- A black-box model offers flexibility but weakens extrapolation arguments and removes parameters useful for control design, digital twins, and health monitoring
- Grey-box dynamic augmentation is the target compromise: preserve the baseline and learn only the missing state dynamics [2]

[TODO: Which industrial benefit should lead: settling prediction, controller design, offline commissioning, or condition monitoring?] [TODO: Instead of health monitoring more focus

on process and root cause analysis, especially the interpretability is waardevol for this.] [TODO: Clearly explain white, black and grey-box] [TODO: Clearly explain interpretability and should be clear in orthogonality section also. lets see where to say what..] [TODO: Decided 2026-10-06: the general definition of interpretability belongs here; Section III states only its operational form (the estimated θ_{base} keeps its physical meaning, which requires a unique estimate) and does not repeat a general one. Candidate sources: Györök et al. [?], Sec. 1 and 3 (“With non-unique θ_b , the baseline model could lose its physical meaning”); Kessels [3], Sec. 5.3.1.3 (negation) and Sec. 6.2 (orthogonality “could enhance physical interpretability”, not yet available beyond linear-in-the-parameter problems), which also states the gap.] ASMPT develops semiconductor assembly and packaging equipment, including high-precision motion systems for advanced packaging. Advanced packaging refers to combining multiple semiconductor functions within a single package, for example through heterogeneous integration of different die types such as logic and memory, and through hybrid bonding, in which dies are directly joined at very fine pitch. [TODO: Shorten to a clause inside the previous sentence (page budget).] Such technologies place strict demands on alignment accuracy, yield, throughput, and manufacturing cost [4], [5]. [TODO: Only partly supported. IRDS covers heterogeneous integration, cost and yield, but never mentions hybrid bonding and its throughput means memory bandwidth. Workman covers sub-micron alignment and yield. Neither supports machine throughput: drop it here or add a source. Workman bib entry: authors incomplete, pages 2071 to 2077 missing.]

This project focuses on a dual-gantry motion system [TODO: Tense and terminology: project becomes paper.] in which two parallel linear actuators support a common beam carrying the payload. The shared beam couples the actuator dynamics, so motion of one actuator influences the other, and the system exhibits position-dependent behaviour and flexible dynamics [1], [6]. [TODO: Teo (abstract only) supports coupling and payload-position dependence, not flexibility. Flexibility rests on Garcia Sec. 2 (flexible joints kb1, kb2): consider flexible joint dynamics.] In such systems, settling behaviour is a key performance metric. After each motion, the system must wait for residual oscillations to decay before the next placement can begin [7], so settling time directly affects throughput. Furthermore, cross-coupling means that the motion of one gantry can disturb the other, making accurate prediction relevant

not only for settling but also for tracking behaviour during simultaneous motion.

Improving settling performance therefore requires an accurate representation of the underlying plant dynamics. ASMPT currently captures part of the position dependence of the dual-gantry system using a baseline model based on frequency-response measurements at multiple operating points with interpolation across the operating space. [TODO: No written source: confirm with Quinten or Jasper. The paper does not compare against this model, so do not imply it.] However, this baseline captures only part of the position dependence and does not systematically account for cross-coupling between the gantries. [TODO: Same confirmation as above.] A more accurate plant model is therefore needed. Such models are not only valuable for settling prediction, but also for digital-twin development, offline evaluation, control design, and condition monitoring. [TODO: Lead with the controller-evaluation digital twin: test a changed controller on the model before the machine (Quinten, 18 Sep). Condition monitoring is not evaluated; its supporting sentence was removed.] Digital twins can support offline simulation of semiconductor manufacturing equipment and help optimize operating parameters before testing changes on the real machine, reducing costly trial-and-error [8]. [TODO: Only partly supported (abstract only). Hwang optimises operating parameters of wafer-processing cluster tools by digital-twin simulation; before testing on the real machine and trial-and-error are not in the source.]

White-box models are attractive because they reflect the physics of the system, which makes them interpretable. First-principles (FP) models are a specific type of white-box model, derived from physical laws and described by quantities such as mass, stiffness, and damping. However, purely physics-based models do not always capture all behaviour needed for accurate prediction across the operating range [9]. For the dual-gantry system, this limitation is also evident in the first-principles model structure of García-Herreros et al. [1], which is only able to capture part of the position-dependent behaviour and cross-coupling. [TODO: Not what Garcia says. Garcia Sec. 2 deliberately neglects cross-arm vibration on the flexible plates, detent forces, and friction variation along the stroke. Our LPV-LFR baseline captures the $M(Y)$ position dependence exactly; the missing dynamics here are the hidden absorber mode and, in the friction twin, Coulomb friction in the truth only (D-204).] [TODO: Proposed replacement of the Garcia sentence: “For the dual-gantry system, the first-principles model of García-Herreros et al. neglects the resonance of the supporting base, the vibration of the cross-arm on its flexible plates, the detent forces of the linear motors, and the variation of friction along the stroke [1, Sec. 2.4].” Verified against Garcia Sec. 2.4, pp. 15 to 16. Our baseline additionally omits Garcia’s Coulomb friction (stated in Sec. II); the simulated stand-ins (absorber, friction) are introduced in Setup.]

Purely data-driven models such as neural networks or Gaussian processes can describe measured behaviour well, but they give less physical insight, are often harder to trust [TODO: Grammar: an and is missing after insight.] at operating

conditions not covered by the training data [9]. Typically these models also require large amounts of data and significant computational power to train. [TODO: Uncited. Large data requirements: Willard Sec. 1. Computational power: no source found, drop or cite. Verified alternative: black-box models spend effort rediscovering dynamics already known from first principles (Hoekstra 2025 EJC p. 1).] For the plant model of the dual-gantry system, both predictive accuracy and physical interpretability are required. However, neither a purely physics-based nor a purely data-driven model is sufficient on its own [9], [10]. [TODO: Cite Willard only: Noel never argues that pure physics models are insufficient. Willard bib: ACM Computing Surveys 2022 (issue 2023), not 2020; month field wrong.] Grey-box modeling is the logical choice, because it is able to combine a physically meaningful baseline model needed for interpretation with data-driven flexibility needed for accurate prediction [10]. [TODO: Noel supports only grey-box combining physical structure with data-driven flexibility. Name model augmentation explicitly and cite Hoekstra 2025 EJC and Hoekstra 2026.] However, related work at ASMPT shows that preserving physical interpretability during model updating remains an open challenge [3]. [TODO: Supported by Kessels thesis Ch. 6 recommendations (PDF p. 222): augmentation can compensate for incorrect FP parameters, compromising their interpretability, and orthogonality methods exist only for linear-in-the-parameter cases. The thesis says industrial wire bonder and never names ASMPT: confirm or reword.]

Learning-based hybrid modeling has already been demonstrated on real industrial systems [11]. [TODO: Supported (Retzler 2024, industrial robot, measured data).] However, a key gap remains: there is no validated grey-box modeling framework for the dual-gantry system that simultaneously addresses coupled multi-input multi-output (MIMO) dynamics, position-dependent behaviour, and preserves physical interpretability. [TODO: Rewrite the gap as a verified intersection. Hoekstra 2025/2026: dynamic LFR augmentation, no LPV scheduling, no closed-loop training. Kessels: closed-loop-trained augmentation on an industrial motion system, but no LFR or LPV, frozen FP parameters, no orthogonality, so closed-loop training alone is not the novelty. Gyro OBC 2026: discrete-time input-output models only; its conclusion names state-space without full-state measurement as future work. Validated must be scoped to simulation.] This project therefore aims to develop and validate such a framework [TODO: Tense and scope: this paper develops and evaluates in simulation.] so that the plant model becomes more accurate for settling prediction while remaining physically meaningful.

B. Research question and scope

- Main question: how should a dual-gantry physics model be augmented to improve settling prediction while retaining physical parameter interpretability?
- Aspect 1 asks whether the payload-position scheduling should remain explicit in a quasi-LPV baseline rather than frozen LTI models

- Aspect 2 asks whether a dynamic parallel augmentation with additional states captures flexible cross-coupling absent from the baseline
- Aspect 3 asks what state-level orthogonality can and cannot guarantee during joint estimation
- Aspect 4 asks whether the model generalises across operating points and motion profiles relative to a black-box state-space model
- Scope separates interpolation, transfer to ASMPT-relevant motion, controller transfer, operating-range extrapolation, and any unavailable hardware evidence

[**TODO:** Does the approved research-plan wording need to remain verbatim, or may the final question reflect the closed-loop and OBC developments?]

The main research question is formulated as follows:

For the dual-gantry system, how should physics-based plant models be augmented to improve parameter interpretability and settling prediction accuracy, particularly of cross-coupling and position-dependent dynamics? [**TODO:** Decision A open: approved wording says improve parameter interpretability, which augmentation cannot do; candidate minimal edit is augmented from closed-loop data to improve settling prediction accuracy while preserving parameter interpretability.]

To answer this research question, the following four aspects are investigated:

1) Position-dependent baseline modeling

In what form should the baseline model be represented for subsequent augmentation, and does position-dependent (LPV) formulation improve prediction compared to position-independent (LTI) alternatives? [**TODO:** LPV versus frozen LTI exists only as a baseline check (lvp_lfr_baseline/scripts/validate_lfr.py), not in the final evaluation; Quinten wants LTI as second baseline.]

2) Augmentation structure

What augmentation structure can effectively capture dynamics not modeled in the physics-based baseline model, in particular flexible cross-coupling dynamics?

3) Interpretability preservation

How can parameter interpretability be maintained during data-driven augmentation, compared to unconstrained approaches? Here, interpretability means that the physical parameters of the baseline retain values close to their physically meaningful initialization after augmentation. [**TODO:** Definition no longer matches the method. The OBC arms start from 10 percent detuned combinations with the prior disabled (D-193) and judge recovery of the ten identifiable combinations. Staying near initialization is what a Lambda penalty trivially gives.]

4) Model generalization

How well does the augmented model generalize to (i) held-out operating points representative of actual industrial operating conditions and (ii) unseen motion profiles relevant to the industrial use context, and how does its

prediction accuracy compare to a black-box state-space model trained on the same data? [**TODO:** Not yet true. The D-196 black-box arm runs on the MA_FRAC 0.10 dataset, not the production ma50 data. Unseen motion profiles partly covered by the Telica cycloidal records (D-206). Controller transfer is not implemented. Align with the validation ladder in Writing/README.md.]

C. Contributions and paper organisation

- Derive a compact continuous-time LPV-LFR realisation of the dual-gantry baseline with rational dependence on payload position
- Extend dynamic parallel LFR augmentation to a closed-loop, self-scheduled state-space setting with learned additional states
- Construct state-level orthogonality for the identifiable physical-parameter tangent space and quantify its true-parameter recovery condition
- Evaluate predictive fit, settling, controller transfer, added-state contribution, interpretability, and black-box performance under matched data
- Paper organisation: baseline, augmentation, OBC, setup, results, discussion, conclusion, then supporting derivations
- Figure plan: no dedicated introduction figure; use the limited figure budget where the baseline, training interconnection, and OBC need visual explanation

[**TODO:** Confirm whether references and appendices are excluded from the 12-page IEEE limit and whether TU/e requires front matter beyond the IEEE paper.]

II. DUAL-DRIVE GANTRY SYSTEM AND PHYSICS BASELINE

A. System and first-principles baseline

The physics baseline of the gantry system used in this paper is the lumped-parameter model of the the dual-drive gantry of García-Herreros et al. [1]. The gantry has two actuators that are connected with a cross-arm and a third actuator that moves the payload along this arm (Fig. 1). [**TODO:** Does this make sense to put here?: Dynamics not represented by the baseline are treated as model discrepancy and addressed by the augmentation in Section III.] Define the actuator forces $u_{\text{act}} = [F_{X1} \ F_{X2} \ F_Y]^T$, the measured positions $q_{\text{act}} = [X_1 \ X_2 \ Y]^T$, and the generalized coordinates $q = [X \ \Theta \ Y]^T$. Let L_b denote the cross-arm length between the two flexible joints. The coordinate and force transformations are

$$q = P^{-T} q_{\text{act}}, \quad u = P u_{\text{act}},$$

$$P = \begin{bmatrix} 1 & 1 & 0 \\ L_b/2 & -L_b/2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (1)$$

where u denotes the generalized force. The generalized coordinates are defined as $X = (X_1 + X_2)/2$ and $\Theta = (X_1 - X_2)/L_b$, while Y is unchanged. These coordinates avoid [**TODO:** This needs to make clear why we choose for this realization. a closed kinematic-chain description and make the coupling caused by

the nonuniform load distribution explicit] [1]. **[TODO: How to connect, nonuniform ... moving payload position]** Moving the payload position Y changes the mass distribution of the gantry and therefore its inertia. This position dependence is retained by representing the baseline in linear parameter-varying (LPV) form, with the measured payload position Y as scheduling variable [12].

Following García-Herreros et al. [1], the baseline uses $\cos \Theta \approx 1$ and $\sin \Theta \approx 0$ and assumes that the Coriolis and centripetal force term is negligible compared with the inertia, damping, and stiffness forces. Under these assumptions, the baseline equations of motion are

$$M(Y)\ddot{q}(t) + C\dot{q}(t) + Kq(t) = Pu_{\text{act}}(t), \quad (2)$$

with

$$M(Y) = \begin{bmatrix} m_1 + m_2 + m_b + m_h & \frac{L_b}{2}(m_1 - m_2) - m_h Y & 0 \\ \frac{L_b}{2}(m_1 - m_2) - m_h Y & J_b + J_h + \frac{L_b^2}{4}(m_1 + m_2) + m_h(d^2 + Y^2) & -m_h d \\ 0 & -m_h d & m_h \end{bmatrix}, \quad (3)$$

$$C = \begin{bmatrix} c_{g1} + c_{g2} & \frac{L_b}{2}(c_{g1} - c_{g2}) & 0 \\ \frac{L_b}{2}(c_{g1} - c_{g2}) & c_{b1} + c_{b2} + \frac{L_b^2}{4}(c_{g1} + c_{g2}) & 0 \\ 0 & 0 & c_y \end{bmatrix}, \quad (4)$$

$$K = \begin{bmatrix} 0 & 0 & 0 \\ 0 & k_{b1} + k_{b2} & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

[TODO: Proposed insertion: “García-Herreros et al. also model Coulomb friction at the three actuators [1, Sec. 2.2]. The baseline omits these terms, so that friction remains a physical effect the augmentation must capture (Section V).” Reason documented in D-204. Second reason? A sign term is not twice differentiable, which [2, Cond. 8] assumes for well-posedness. because: ?] [TODO: Look at this section, the above state a lot, look fine] The masses, flexible-joint stiffness and damping coefficients, and **[TODO: geometric dimensions]** are shown in Fig. 1. Here, J_b and J_h denote the yaw inertias, while c_{g1} , c_{g2} , and c_y denote the actuator damping coefficients. The nominal values are taken from [1, Table I] and listed in Appendix A-B **[TODO: Not sure if to add this, depends on**

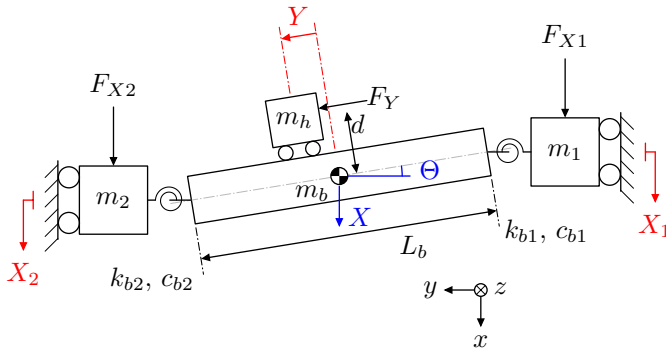


Fig. 1: Lumped-parameter gantry schematic adapted from the mechanical arrangement in [1]. Red labels denote actuator coordinates and blue labels generalized coordinates.

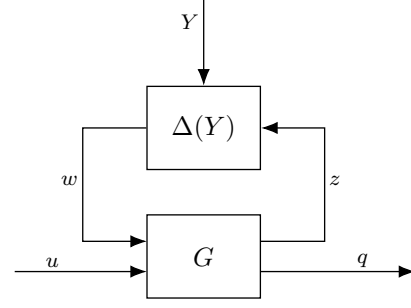


Fig. 2: Self-scheduled **baseline** interconnection. The **Delta-block** closes the **latent variables** of G through $w = \Delta(Y)z$, with $\Delta(Y) = YI_6$ and Y a component of the **state**.

real data or not; they were not identified on the present Telica machine.] To prevent the definition of q from being parameter-dependent, the cross-arm length L_b is fixed at the nominal value listed in Appendix A-B rather than treated as a free model parameter. **[TODO: Determine and justify the operating range of Y ; distinguish the mechanical stroke, simulated-data range, and range covered by the measured ILC records.]**

B. Self-scheduled LPV-LFR representation

- Y is a state and the scheduling variable; $M(Y)^{-1}$ makes the model rational in Y (Drenth)
- $M(Y) = M_0 + YM_1 + Y^2M_2$; Y enters only through $Y\ddot{q}$ and $Y^2\ddot{q}$, realized by repeating Y (this work)
- Latent variables z, w and Delta-block $\Delta(Y) = YI_6$; not claimed minimal (this work)
- LTI part G in the continuous-time form of Drenth Eq. (6); its first block row is the state equation (adapted)
- Closing the algebraic loop gives back (2) exactly (this work)
- Well posed iff $M(Y)$ nonsingular, true for every Y ; closed-form $M(Y)^{-1}$, no numerical solve (Drenth Def. 1; this work)

The measured payload position Y is both a state and the scheduling variable, making the baseline a self-scheduled LPV model **[TODO: source, or do i state it earlier..?]**. Solving (2) for $\ddot{q}$ requires $M(Y)^{-1}$, so the model is rationally dependent on Y . An Linear Fractional Representation (LFR) is introduced to separate this rational scheduling dependence. We move the dependence out of the LTI part G and into a Delta-block $\Delta(Y)$, with the scheduling being connected in feedback with G (Fig. 2) [2], [13].

To separate the scheduling variable Y , deconstruct the **inertia matrix** (3) into its Y dependent components,

$$M(Y) = M_0 + YM_1 + Y^2M_2. \quad (5)$$

M_0 , M_1 and M_2 are given in Appendix A-A. Substituting (5) into (2) gives

$$M_0\ddot{q} + M_1(Y\ddot{q}) + M_2(Y^2\ddot{q}) = u - C\dot{q} - Kq. \quad (6)$$

Here Y only enters through $Y\ddot{q}$ and $Y^2\ddot{q}$, with $Y^2\ddot{q} = Y(Y\ddot{q})$. This lets us apply the same scheduling variable twice instead of introducing a second scheduling variable Y^2 ,

$$\ddot{q} \xrightarrow{Y} a_1 \xrightarrow{Y} a_2, \quad (7)$$

with $a_1 = Y\ddot{q}$ and $a_2 = Y^2\ddot{q}$.

[TODO: Left here, ...] We treat a_1 and a_2 as independent signals (6) becomes

$$M_0\ddot{q} + M_1a_1 + M_2a_2 = u - C\dot{q} - Kq, \quad (8)$$

which contains no Y . The **latent variables** are $z = \text{col}(\ddot{q}, a_1)$ and $w = \text{col}(a_1, a_2)$, with the **Delta-block** mapping z to w .

$$w = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = \begin{bmatrix} Y\ddot{q} \\ Y^2\ddot{q} \end{bmatrix} = \underbrace{\begin{bmatrix} YI_3 & 0 \\ 0 & YI_3 \end{bmatrix}}_{\Delta(Y)=YI_6} \begin{bmatrix} \ddot{q} \\ a_1 \end{bmatrix} = \Delta(Y)z. \quad (9)$$

We do not claim this **realization** to be **minimal** **[TODO: I have an SVD reduction add this in the appendix or not?]**. We can write the LTI part G of the LFR in the continuous-time form, [13, Eq. (6)],

$$\begin{bmatrix} \dot{\tilde{x}} \\ z \\ q \end{bmatrix} = G \begin{bmatrix} \tilde{x} \\ w \\ u \end{bmatrix}, \quad (10)$$

where $\tilde{x} = \text{col}(q, \dot{q})$ is the **state** and q the **output**. The first block row is the **state equation**, with $\ddot{q}$ obtained from (8) by multiplying with M_0^{-1} . For the blocks of G Appendix A-A. Closing the **algebraic loop** with (9) results in $a_1 = Y\ddot{q}$ and $a_2 = Y^2\ddot{q}$, which turns (8) back into (2). The LFR is therefore an exact realization of the baseline. The LFR is **well posed** if the **algebraic loop** has a unique solution for every Y [13, Def. 1]. For the gantry realization we show that this conditions holds when $M(Y)$ is nonsingular (Appendix A-A). For positive **masses** and **inertias**, $M(Y) \succ 0$ for every $Y \in \mathbb{R}$ (Appendix A-C). Therefore the **LFR** is well posed over the whole Y -operating range.

C. Physical parameters and identifiable combinations

- The fourteen raw parameters enter (2) only through the ten combinations θ_{base} , the gantry counterpart of base parameters (this work; Gautier and Khalil)
- Exactly ten independent functions of θ are structurally identifiable, namely θ_{base} ; proof from (2) alone by counting matrix entries (this work; definition Hong, Ovchinnikov)
- Four lost directions; estimates reported in θ_{base} ; recovery from the records in Setup
- Positive θ_{base} does not give $M(Y) \succ 0$; that holds iff (12) (this work; proof Appendix A-C)
- Training keeps (12) at every iterate; the map and its justification are in Section III-C

The fourteen raw parameters θ of Table I enter (2) only through

$$\theta_{\text{base}} = [k_{b,\Sigma}, c_{g1}, c_{g2}, c_y, c_{b,\Sigma}, m_h, m_\Sigma, m_\Delta, J_{\text{eff}}, d]^\top, \quad (11)$$

with $k_{b,\Sigma} = k_{b1} + k_{b2}$, $c_{b,\Sigma} = c_{b1} + c_{b2}$, $m_\Sigma = m_1 + m_2 + m_b$, $m_\Delta = m_1 - m_2$ and $J_{\text{eff}} = J_b + J_h + \frac{L_b^2}{4}(m_1 + m_2)$, the gantry counterpart of base inertial parameters [14].

Proposition 1: Exactly ten independent functions of θ are structurally identifiable [15, Def. 2.5], [16, Def. 7] from (2), namely θ_{base} .

Proof: Two parameter vectors with the same input-output behaviour satisfy

$$(M(Y) - M'(Y))\ddot{q} + (C - C')\dot{q} + (K - K')q = 0$$

along every trajectory. Every $(q, \dot{q}, \ddot{q})$ is attained, so their matrices coincide. Conversely, equal matrices give equal behaviour. Up to symmetry, M_0, M_1, M_2, C, K have twelve nonzero entries. Three of them, $M_{0,33}$, $-M_{1,12}$ and $M_{2,22}$, equal m_h , which leaves ten distinct functions of θ . With respect to $(m_h, d, m_1, m_b, J_b, c_{g1}, c_{g2}, c_{b1}, c_y, k_{b1})$, their Jacobian is block triangular with determinant $m_h L_b^2/2 \neq 0$. ■

The four lost directions are $k_{b1} - k_{b2}$, $c_{b1} - c_{b2}$, $J_b - J_h$ and a mass transfer from the carriages to the cross-arm, so estimates are reported in θ_{base} . Recovery from the closed-loop records is examined in Section V.

Positive θ_{base} does not imply $M(Y) \succ 0$. For $m_h, m_\Sigma, J_{\text{eff}} > 0$, $M(Y) \succ 0$ for every Y if and only if

$$m_\Sigma J_{\text{eff}} > \frac{L_b^2}{4} m_\Delta^2 \quad (12)$$

(Appendix A-C). Training therefore uses logarithmic coordinates for the nine positive combinations and a tanh-bounded m_Δ (Section III-C), so (12) and well-posedness hold at every iterate.

III. DYNAMIC LPV-LFR AUGMENTATION

- The section extends the baseline with learned dynamics and identifies the result from closed-loop records; it names its three subsections (style profile)
- Contribution share: the realisation on the self-scheduled gantry baseline with joint estimation of θ_{base} , and its identification in closed loop with the known controller (MUST ESTABLISH)

The baseline of Section II-B is augmented with learned dynamics and identified from records measured under feedback. Section III-A defines the augmented model, Section III-B the initial state of each training window, and Section III-C the closed-loop identification criterion. This work contributes the realisation of the augmentation on the self-scheduled gantry baseline. The physical parameters are estimated jointly with the augmentation, in the combinations of Section II-C, and the model is identified in closed loop with the known controller. **[TODO: VERIFY keys: hoekstra2025lfr = EJC 86 (2025) 101304, hoekstra2026lfr = arXiv:2602.17297 (submitted to Automatica). refs.bib and docs/references.md rows 46 and 130 now agree.]** **[TODO: Missing bib entries: gyorok2026obc (IFAC J. Systems and Control 35:100376, 2026, arXiv 2511.01321) and forsell1999revisited (Automatica 35(7), pp. 1215 to 1241, 1999). Only the 1998 LiU report of Forsell and Ljung is held locally; check that its Sec. 5.2 matches the Automatica numbering.]**

A. Augmented model

- If the omitted effects are dynamic, the six baseline states cannot represent them; refitting θ_{base} or a static augmentation adds no states (EJC Sec. 2)
- The model is Hoekstra's LFR augmentation structure with a fixed interconnection realising S-DP; its baseline block is the LFR of Section II (EJC Sec. 3, 2026 Table 1)
- Parallel because the orthogonal-by-construction method of Section IV is defined for the additive structure (Györök 2026); the baseline stays an explicit term
- State augmentation only, output map not learned (?)
- f_{base} is one RK4 step with the input held and $M(Y)$ per stage, so it stays self-scheduled; affine normalisation from the training records, unlike Hoekstra Eq. (28)
- The learned correction is a discrete-time correction after the step (?)
- One feedforward network with tanh hidden layers supplies f_{aug} and g_{aug} ; plain feedforward suffices for a parallel structure (EJC Sec. 4.3)
- The correction acts on all physical rows, X and Y included; on these axes it meets no stiffness (?: why the position rows)
- The network depends on Y through the state but has no LPV structure, unlike Drenth Eq. (5.2) (?)
- The interconnection is acyclic, so it adds no algebraic loop; the baseline loop is well posed by Section II
- A zero output layer reproduces the baseline from the same physical initial state (2026 Eq. (29)); additional states have no physical coordinates
- The baseline/network split is not unique (2026 Sec. 5.2); interpretable means the estimated θ_{base} keeps its physical meaning (Györök Sec. 3); Section IV makes the estimate unique

If the effects that the baseline omits are dynamic, its six states cannot represent them. Refitting θ_{base} adds no states. Neither does a static augmentation, which corrects the state update of the baseline without extending its state [17, Sec. 2]. The augmentation therefore needs states of its own.

We use the LFR-based augmentation structure of Hoekstra et al. [2], [17], in which a fixed interconnection connects a baseline block and a learning block [17, Sec. 3.2]. The baseline block is the self-scheduled LFR of Section II-B, so the augmented model is an LFR whose baseline block is itself an LFR in Y . The interconnection realises the dynamic-parallel structure [2, Table 1]

$$\begin{aligned}\tilde{x}_{k+1} &= f_{\text{base}}(\theta_{\text{base}}, \tilde{x}_k, u_k) + f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_k), \\ \bar{x}_{k+1} &= g_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_k), \\ \hat{y}_k &= h_{\text{base}}(\tilde{x}_k),\end{aligned}\quad (13)$$

where $\tilde{x} = \text{col}(q, \dot{q})$ is the physical state of Section II-B, $\bar{x} \in \mathbb{R}^{n_{\bar{x}}}$ the additional state, $\hat{x} = \text{col}(\tilde{x}, \bar{x})$, $u = Pu_{\text{act}}$ the generalised force of (1), and $h_{\text{base}}(\tilde{x}) = P^{\top}q$ the measured positions q_{act} . We choose the parallel interconnection because the orthogonal-by-construction parametrisation that Section IV extends is defined for this additive structure [?]. It also keeps

the baseline as an explicit term of the state update. The output map is not augmented. **[TODO: Missing: reason for state augmentation only ($h_{\text{aug}} = 0$). Candidate: h_{base} is the exact kinematic map of the measured positions, the premise under which Kessels initialises positions from the output [3, Remark 5.3]; an output augmentation depending on the input would also close an algebraic loop with the controller [3, Remark 5.1].]**

The baseline transition f_{base} is one fourth-order Runge-Kutta (RK4) step of the equations of motion (2) over the sample period T_s , with the input held over the step. Each stage evaluates $M(Y)$ at its own state, so f_{base} remains self-scheduled. The step acts on signals normalised with statistics of the training records,

$$\tilde{x} = T_x(\text{col}(q, \dot{q}) - \mu_x), \quad T_x = \text{diag}(\sigma_x)^{-1}, \quad (14)$$

where μ_x and σ_x are the mean and standard deviation of each state channel. The actuator forces u_{act} and positions q_{act} are normalised in the same way. Hoekstra et al. only scale the signals and take σ_x from a simulation of the baseline [2, Sec. 5.3]. Here the normalisation is affine, and the state statistics come from the measured positions $q = P^{-\top}q_{\text{act}}$ and their first differences. The learned correction acts on the result of the RK4 step. f_{aug} is therefore a discrete-time correction at the sample rate, whereas f_{base} integrates the continuous-time physics. **[TODO: Missing: reason for correcting the state after the RK4 step rather than adding a learned force inside the continuous-time derivative. No decision entry gives a reason other than gradient magnitude.]**

One multilayer perceptron with tanh hidden layers and a linear output layer supplies both learned functions,

$$\begin{bmatrix} f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_k) \\ g_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_k) \end{bmatrix} = \phi_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_k), \quad (15)$$

where the first six outputs form f_{aug} and the last $n_{\bar{x}}$ form g_{aug} (Fig. 3). A plain feedforward network suffices because the baseline is augmented additively, whereas a series structure needs a residual network to reproduce the baseline at initialisation [17, Sec. 4.3]. As in the dynamic-parallel structure, f_{aug} corrects every physical row [17, Sec. 2], so the learned component can act on the X and Y axes. On these axes a correction meets no stiffness ((4)), so a persistent correction accumulates in position. **[TODO: Missing: why the position rows are corrected rather than kept as the integrals of the velocities. Context: Hoekstra et al. detach the free integrators of their vehicle and learn only the input-to-velocity map, which needs measured velocities [2, Sec. 7]; here only positions are measured. D-103 requires only that X and Y are reachable.]** The network depends on Y through $\hat{x}$, so it can represent position-dependent corrections, but unlike the augmented LPV-LFR of Drenth [18, Eq. (5.2)] it has no LPV structure. **[TODO: Missing: reason for an unrestricted network rather than an LPV-structured augmentation. The research plan (Aspect 2) proposed the interconnection in LPV-LFR form; this is a departure.]** No learned function feeds the baseline block or itself within a step, so the interconnection graph is acyclic and adds no algebraic

The physical rows start from the baseline, following the model-based initialisation of Hoekstra et al. [20, Sec. 3.1]. For zero input, every rest state with $\Theta = 0$ is an equilibrium of (2), because K in (4) vanishes on X and Y . We linearise the baseline at rest at $Y = 0$ and discretise it by zero-order hold over T_s , which gives (A_d, B_d, C_d) in normalised coordinates.

The reconstructability map of this model [20, Eqs. (15) to (17)] gives

$$W^b = [A_d^n O_n^+ \quad r_n - A_d^n O_n^+ T_n], \quad (19)$$

where O_n is the observability matrix of (A_d, C_d) over $n + 1$ samples, T_n the Toeplitz matrix of its Markov parameters, r_n the input-to-state map over the same samples, and the two blocks act on the output and input parts of ξ_τ . The encoder paper assumes a stable baseline. Ours is marginally stable, but the measured positions make (A_d, C_d) observable, so O_n has full column rank and the map exists. Unlike [2, Eq. (30)], which fits the baseline encoder to simulated baseline states, we compute the map directly. The model-based initialisation gives better starting values and faster convergence than a random encoder, at comparable final accuracy [20, Sec. 4.4]. The encoder is trained jointly with the model, so the linearisation only sets its starting value. **[TODO: Missing: why linearise at $Y = 0$. The encoder paper also linearises “around the 0 point” [20, Sec. 4.2] but gives no reason.]** **[TODO: Missing: disclosure. W^b is built from the nominal parameters, while θ_{base} starts detuned and is co-estimated, which the encoder paper leaves out of scope [20, footnote 1]. State which prior information this uses.]** **[TODO: Missing: why the model-based rather than the data-based initialisation [20, Sec. 3.2]. Candidate: negligible computation [20, Sec. 4.4].]**

C. Closed-loop identification

- The records are measured under feedback and the model must predict the machine under feedback, so we fit the closed-loop response, not a rollout driven by the recorded input (2026 Eq. (22))
- Inspired by Kessels (Eqs. (5.12), (5.13); Remark 5.6); for linear systems the indirect approach, where an output-error criterion remains consistent (Forssell and Ljung); a motivation, not a guarantee
- X and Y are free, so the open-loop gantry violates the stability condition behind SUBNET consistency (Beintema Cond. 1); the loop with the controller can satisfy it
- Both loops share r , u^{ff} and the controller; subtracting them gives our residual form, exact under a compatible controller, signals and initial state (check in Setup)
- $x_\tau^c = 0$ needs no controller initialisation and matches the encoder; Kessels’ direct form needs the machine’s controller state (Remark 5.4)
- No plant feedthrough fixes the step order: no algebraic loop, no added delay
- θ_{base} , θ_{aug} and η minimise the closed-loop output error over every window sample, controller fixed
- Moved from Section II-C: state and justify the training coordinates of θ_{base} here. Nine positive combinations in logarithmic coordinates; the signed m_Δ through

$$m_\Delta = \rho \frac{2}{L_b} \sqrt{m_\Sigma J_{\text{eff}}} \tanh(\eta_0 + s \xi_\Delta) \quad (20)$$

Justification: every ξ_Δ satisfies (12), so $M(Y) \succ 0$ and the LFR stays well posed at every optimiser iterate without clamp or penalty (D-229); constraint by parameterization

as in [21]; it guarantees positive-definite inertia, not positive raw masses (positive m_1, m_2 need $|m_\Delta| < m_\Sigma - m_b$, with m_b unidentifiable); η_0 , s in Appendix A-C, ρ in Setup. Notation clash: η_0 , s , ξ_Δ against η , s_k , ξ_τ of this section (?)

- The horizons are separate quantities with values in Setup; a closed-loop fit can hide plant error (Results)

The records are measured under feedback, and the model must predict the machine under feedback: its response to a reference, a feedforward and a known controller. We therefore identify the model from this closed-loop response, rather than from a simulation driven by the recorded input as in [2, Eq. (22)]. Kessels trains augmented models in this way, simulating each window in closed loop with the known controller [3, Eqs. (5.12), (5.13)]. He also reports that identifying the open-loop model from his closed-loop data failed [3, Remark 5.6]. For linear systems, fitting the closed loop from the reference with a known controller is the indirect approach to closed-loop identification [2, Sec. 5.2]. The reference is uncorrelated with the noise, so an output-error criterion remains consistent under the conditions given there. For our nonlinear model, this is a motivation, not a guarantee.

The free axes of the baseline give a second reason. X and Y have no stiffness in (4), so the open-loop gantry is marginally stable: a difference in initial velocity leaves a permanent position offset. The consistency of the identification method [2, Sec. 5.5] rests on that of SUBNET, which assumes an incrementally exponentially output stable data-generating system [19, Cond. 1]. The open-loop gantry does not satisfy this condition, whereas the loop with the stabilising controller can. Kessels uses the same principle to update a model of a system that is unstable in open loop [3, Sec. 5.1]. For our nonlinear model, this makes the condition attainable but does not prove consistency. **[TODO: Missing: evidence on the thesis data that open-loop rollouts accumulate position error on X and Y ; measure and report in Results.]**

Both loops are driven by the same reference r and feed-forward u^{ff} through the same controller, and differ only in the output they feed back (Fig. 4). The controller of each record is known, linear and time invariant, with state-space matrices $(A_{\text{fb}}, B_{\text{fb}}, C_{\text{fb}}, D_{\text{fb}})$ mapping the position error in actuator coordinates to actuator forces. For the machine, with controller state s , and the model, with controller state $\hat{s}$, the loops are

$$\begin{aligned} s_{k+1} &= A_{\text{fb}} s_k + B_{\text{fb}} (r_k - y_k), \\ u_{\text{act},k} &= C_{\text{fb}} s_k + D_{\text{fb}} (r_k - y_k) + u_k^{\text{ff}}, \\ \hat{s}_{k+1} &= A_{\text{fb}} \hat{s}_k + B_{\text{fb}} (r_k - \hat{y}_k), \\ \hat{u}_{\text{act},k} &= C_{\text{fb}} \hat{s}_k + D_{\text{fb}} (r_k - \hat{y}_k) + u_k^{\text{ff}}. \end{aligned} \quad (21)$$

We implement the model loop by subtracting the machine’s equations from the model’s. This removes r and u^{ff} and gives

$$\begin{aligned} x_{k+1}^c &= A_{\text{fb}} x_k^c + B_{\text{fb}} (y_k - \hat{y}_k), & x_\tau^c &= 0, \\ \hat{u}_{\text{act},k} &= u_{\text{act},k} + C_{\text{fb}} x_k^c + D_{\text{fb}} (y_k - \hat{y}_k), \end{aligned} \quad (22)$$

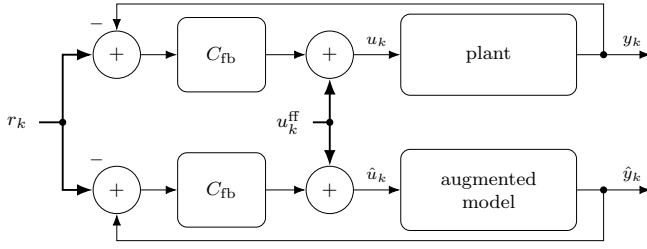


Fig. 4: Recorded plant loop (top) and augmented-model loop (bottom), driven by one shared reference r_k and feedforward u_k^{ff} , with the same feedback controller. The model output is evaluated before the input advances its state to the next sample.

where $x^c = \hat{s} - s$ is the difference of the controller states. The model thus receives the recorded input plus the controller's response to its own output error, and $\hat{u}_k = P\hat{u}_{\text{act},k}$ replaces u_k in (13). The two forms are equal when the recorded input was produced by this controller from the same output samples and initial state. **[TODO: Check for Setup: the training loop uses the controller re-discretised at the model rate, while the records were generated at a higher rate. Run the re-discretised controller on the recorded $r - y$ and compare its feedback force with the recorded one. A rollout with $\hat{y} = y$ gives $\hat{u} = u$ by construction and checks nothing.]**

Because x^c is a difference of controller states, it needs no initialisation. Setting $x_\tau^c = 0$ at every window start places the model's controller in the machine's controller state, $\hat{s}_\tau = s_\tau$, just as the encoder places the model state on the recorded trajectory. Kessels' direct form instead needs the machine's controller state at every window start, read from the machine or reconstructed from the reference, the output and the controller [3, Remark 5.4]. Model error made before τ is not carried into the window, which matches the truncated objective below. The free-run validation of Section V simulates whole records and does carry it. The augmented model has no feedthrough, so each sample is evaluated in a fixed order: $\hat{y}_k$ from $\tilde{x}_k$, then $\hat{u}_{\text{act},k}$ from (22), then the model and controller states advance. The controller feedthrough D_{fb} therefore closes no algebraic loop, and the loop adds no delay.

Over the window starts $\mathcal{T}$ and the window length n_f , the parameters $\vartheta = (\theta_{\text{base}}, \theta_{\text{aug}}, \eta)$ minimise the closed-loop output error

$$V(\vartheta) = \frac{1}{|\mathcal{T}| n_f n_y} \sum_{\tau \in \mathcal{T}} \sum_{\ell=0}^{n_f-1} \|y_{\tau+\ell} - \hat{y}_{\tau+\ell|\tau}\|_2^2 \quad (23)$$

subject to (13), (17) and (22), where $n_y = 3$ and the outputs are normalised. As in [2, Eq. (22)], every sample of a window is scored. θ_{base} is estimated in the combinations of Section II-C, jointly with the network and the encoder, while the controller stays fixed. Kessels' criterion, in contrast, leaves the first-principles parameters fixed [3, Eq. (5.12)]. The encoder history n , the window length n_f and the free-run validation horizon are separate quantities, whose values are given in Section V. A good closed-loop fit does not by itself show that the model

is correct in open loop, because the controller acts on the output error and can mask plant error. Section VI therefore also examines the model outside the training loop.

IV. INTERPRETABILITY BY ORTHOGONAL CONSTRUCTION

This section extends Györök et al. their orthogonal-by-contruction (OBC) method for linear input-output models to nonlinear state-space models and applies it to the gantry model **[TODO: Source]**.

A. Negation in the Gantry State Equation

[TODO: Subsection title improve.] Fitting the physical parameters θ_{base} of the baseline model simultaneously with the ANN weights θ_{aug} during model training allows the ANN to represent dynamics which could have been explained by updating the baseline parameters. The ANN correction is additive to the baseline state update, with corrections to the physical states f_{aug} and the additional states g_{aug} .

This one-step correction enables the ANN to

The additive correction of the ANN can realise the same one-step

Joint estimation fits the physical parameters θ_{base} and ANN weights θ_{aug} together. The ANN writes f_{aug} into the six physical rows and g_{aug} into the additional rows. Their additive connection permits different parameter pairs to realise the same one-step map [?, Ex. 2], [22, Ex. 1]. The physical parameter directions are the columns of the transition sensitivity [22, Eq. (15)]

$$\Phi_{\bar{\theta}}(\tilde{x}, u) = \left. \frac{\partial f_{\text{base}}(\theta_{\text{base}}, \tilde{x}, u)}{\partial \theta_{\text{base}}} \right|_{\theta_{\text{base}} = \bar{\theta}_{\text{base}}} \in \mathbb{R}^{6 \times 10}, \quad (24)$$

at the expansion point $\bar{\theta}_{\text{base}}$. A learned write can cancel a physical displacement $\delta\theta$ to first order:

$$\begin{aligned} f_{\text{base}}(\bar{\theta}_{\text{base}} + \delta\theta, \tilde{x}, u) + [f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}, u) - \Phi_{\bar{\theta}}(\tilde{x}, u) \delta\theta] \\ \approx f_{\text{base}}(\bar{\theta}_{\text{base}}, \tilde{x}, u) + f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}, u). \end{aligned} \quad (25)$$

The data cannot attribute this direction to either component. We call this negation and aim for a split whose physical estimate is not selected by the initial parameter pair. **[TODO: From Section III (2026-10-06): this section owns the term negation. Cite Kessels' definition [3, Sec. 5.3.1.3] (learned terms that "(partially) negate the FP terms") and state that the first-order cancellation above makes it precise for the parameter directions; his notion is broader. Section III ends with "Section IV extends the method so that this estimate is unique", so open from that uniqueness requirement.]**

For the gantry, differentiating (2) with respect to its j -th identifiable combination gives

$$\frac{\partial \ddot{q}}{\partial \theta_{\text{base},j}} = -M(Y)^{-1} \left[\frac{\partial M(Y)}{\partial \theta_{\text{base},j}} \ddot{q} + \frac{\partial C}{\partial \theta_{\text{base},j}} \dot{q} + \frac{\partial K}{\partial \theta_{\text{base},j}} q \right]. \quad (26)$$

The RK4 integrator propagates this acceleration sensitivity into the normalised one-step transition. The mass combinations act through $\ddot{q}$, the damping combinations through $\dot{q}$, and the stiffness combination through q . Hence, a negating write

acts like an added mass, damping, or stiffness force. The factor $M(Y)^{-1}$ makes these directions depend on the **payload position**.

The fourteen **raw parameters** contain four **structural null directions**, whereas the ten **combinations** θ_{base} of Section II-C remove that ambiguity. **OBC** therefore protects the ten **combination directions**. Recovery of their true values requires the additional condition in Section IV-C.

B. Orthogonal-by-Construction for the Gantry

For an **input-output baseline** linear in its parameters, Györök et al. fit the stacked **ANN output** to the **baseline regressor** and subtract that fit:

$$\begin{aligned}\theta_{\text{aux}} &= (\Phi^\top \Phi)^{-1} \Phi^\top F_{\theta_a}^{\text{ANN}}, \\ \tilde{F}_{\theta_a}^{\text{ANN}} &= F_{\theta_a}^{\text{ANN}} - \Phi \theta_{\text{aux}}, \\ 0 &= \Phi^\top \tilde{F}_{\theta_a}^{\text{ANN}}, \\ \hat{y}_k &= \phi(x_k) \theta_b + f_{\theta_a}^{\text{ANN}}(x_k) - \phi(x_k) \theta_{\text{aux}},\end{aligned}\quad (27)$$

[?, Eqs. (8), (10), (11), (13)]. The **corrected stack** is orthogonal to the **baseline range** without the **penalty weight** required by [22].

We apply (27) to the **physical rows** of the **normalised RK4 state transition**. Its **regressor** is (24), because f_{base} is rational in θ_{base} through $M(Y)^{-1}$ and has no exact **linear-in-parameters form**. Unlike [22, Eq. (18)], the **affine offset** of this expansion is not protected, because it multiplies no **estimated parameter**. **[TODO: Gamma excluded: the protected set is the range of Phi (parameter uniqueness, [?, Eq. (39)]), not the range of [Phi Gamma] (output overlap, [22, Eq. (18)]). Confirm with supervisors, since D-190 records their condition as non-overlap with the baseline output.] [TODO: Should the thesis call the affine arm a stricter non-negation definition or only a sensitivity analysis around the chosen tangent definition?] [Proposed answer: neither; the affine arm is removed from the campaign (RESULTS-DESIGN change 30), exclusion of Γ stated with the reason above. The two affine runs (84037, 84034) are incomplete and are not quoted.] [Add refs.bib entry gyorok2026obc (arXiv 2511.01321v3); hoekstra2026lfraug does not exist in refs.bib.]**

Only **positions** are measured, while an **open-loop baseline reconstruction** drifts on the **free axes** X and Y . We therefore build fixed **reference tuples** from every valid **training sample**:

$$\begin{aligned}q_i &= P^{-\top} y_i, & \dot{q}_i &= \frac{-q_{i+2} + 8q_{i+1} - 8q_{i-1} + q_{i-2}}{12 T_s}, \\ \tilde{x}_i &= \text{col}(q_i, \dot{q}_i), & \bar{x}_i &= 0,\end{aligned}\quad (28)$$

where $u_i = P u_{\text{act},i}$ and $i = 1, \dots, N$. The implementation stores the recorded **actuator force** $u_{\text{act},i}$ and applies P inside the **physical block**. The **stencil** does not cross **record boundaries**. Each **tuple** uses its measured **scheduling value** Y_i , and the **recorded input** already contains the **feedback action**. The **tuples** are normalised before the following **stacks** are evaluated. A fixed **auxiliary evaluation set** is permitted by [?].

On these **tuples**, the construction (27) becomes

$$\begin{aligned}\Phi_{\tilde{\theta}}(\tilde{X}, U) &= \text{col}(\Phi_{\tilde{\theta}}(\tilde{x}_1, u_1), \dots, \Phi_{\tilde{\theta}}(\tilde{x}_N, u_N)) \in \mathbb{R}^{6N \times 10}, \\ F_{\text{aug}} &= \text{col}(f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_1, u_1), \dots, f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_N, u_N)), \\ \theta_{\text{aux}} &= \Phi_{\tilde{\theta}}(\tilde{X}, U)^+ F_{\text{aug}}, \\ 0 &= \Phi_{\tilde{\theta}}(\tilde{X}, U)^\top (F_{\text{aug}} - \Phi_{\tilde{\theta}}(\tilde{X}, U) \theta_{\text{aux}}), \\ 0 &= \Phi_{\tilde{\theta}}(\tilde{X}, U)^\top \frac{\partial (F_{\text{aug}} - \Phi_{\tilde{\theta}}(\tilde{X}, U) \theta_{\text{aux}})}{\partial \theta_{\text{aug}}},\end{aligned}\quad (29)$$

with $\hat{x}_i = \text{col}(\tilde{x}_i, \bar{x}_i)$. The **protected range** contains every first-order change of the stacked **physical transition** generated by θ_{base} . **Full column rank** is required and monitored at every rebuild; the thesis-data **rank** is reported in Section VI. The fourth line of (29) is [?, Lemma 3]. Differentiating that **stacked identity** gives the last line because the **basis** is fixed with respect to θ_{aug} .

Fitting the **coefficient** on the **rollout** itself would make the **correction** depend on later **rollout states**. We instead compute θ_{aux} once per **objective** from the fixed **tuples** and differentiate through this solve. The **sensitivity** and $\tilde{\theta}_{\text{base}}$ are detached within an **epoch** and rebuilt between **epochs** at the current θ_{base} (D-192). The **corrected augmented model** is

$$\begin{bmatrix} \tilde{x}_{k+1} \\ \bar{x}_{k+1} \end{bmatrix} = \begin{bmatrix} f_{\text{base}}(\theta_{\text{base}}, \tilde{x}_k, u_k) \\ 0 \end{bmatrix} + \begin{bmatrix} f_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_k) - \Phi_{\tilde{\theta}}(\tilde{x}_k, u_k) \theta_{\text{aux}} \\ g_{\text{aug}}(\theta_{\text{aug}}, \hat{x}_k, u_k) \end{bmatrix}. \quad (30)$$

At **rollout point** $(\tilde{x}_k, u_k)$, the **correction** uses the frozen **expansion point** and the current **objective's coefficient**, as in [?, Eq. (13)]. A **hand-written forward sensitivity** propagates the required **directional derivative** through **RK4** (appendix). The zero **baseline rows** leave g_{aug} uncorrected. **[Figure (new): geometric decomposition of F_{aug} into its part in range $\Phi_{\tilde{\theta}}$ and its orthogonal part, on the reference stack at $\tilde{\theta}_{\text{base}}$, with a curved surface marking that the tangent is local. Inset: state partition showing the correction on the physical rows, uncorrected added-state rows, and their delayed influence through the recursion rather than a direct output path.] [TODO: Is one-step state orthogonality the intended thesis contribution, or must the trajectory-level penalty also appear despite its unresolved weight selection?]**

C. Scope and Condition for True Parameter Recovery

The **guarantee** is local in θ_{base} and holds for the **one-step stack** over the chosen **tuples**. It uses the **Euclidean inner product** $\langle a, b \rangle = a^\top b$ in **normalised state coordinates**, so the **state normalisation** defines its **metric**. It is aggregate rather than pointwise and does not establish **global uniqueness**. The **tuples** lie on $\bar{x} = 0$, so the **coefficient** does not see directions that act only through nonzero **added states**. The **added-state rows** are never corrected, and g_{aug} can affect the **physical rows** one step later through f_{aug} . The **rollout correction** depends only on $(\tilde{x}_k, u_k)$, and there is no **stacked guarantee** away from the $\bar{x} = 0$ **slice**. Consequently, the construction does

not imply trajectory-output orthogonality. The consistency and zero-covariance results of [?, Thms. 16, 18] do not transfer.

Algebraic non-overlap does not imply the true split. Let θ_{base}^* be the true parameters and Δ^* the true one-step discrepancy on the tuples, in the same normalised coordinates,

$$\Delta^* = \text{col}(\tilde{x}_{i+1} - f_{\text{base}}(\theta_{\text{base}}^*, \tilde{x}_i, u_i))_{i=1}^N, \quad (31)$$

where $\tilde{x}_{i+1}$ is the recorded successor, built with the same stencil. Suppose the data are noise-free, f_{base} is affine in θ_{base} near θ_{base}^* , the stack Φ of (29) is evaluated at θ_{base}^* , and the parameters minimise the one-step least-squares error on the tuples. The corrected learned write then lies in the orthogonal complement of range Φ , and the parameter error is

$$\hat{\theta}_{\text{base}} - \theta_{\text{base}}^* = \Phi^+ \Delta^*, \quad \hat{\theta}_{\text{base}} = \theta_{\text{base}}^* \iff \Phi^T \Delta^* = 0, \quad (32)$$

which is the state-level counterpart of [?, Cond. 4, Eq. (21)]. The training objective is a closed-loop multistep output error, so (32) is a one-step surrogate, not the bias of the trained model. For the gantry equations, no nonzero missing generalized force is orthogonal to the sensitivity at a single tuple. The contributions must therefore cancel over the reference stack. Such aggregate cancellation requires designed excitation [?, Remark 5].

Two overlap measures follow from the same inner product. For a stacked vector v over the tuples, with Φ the stack at $\bar{\theta}_{\text{base}}$ and $F_{\text{base}} = \text{col}(f_{\text{base}}(\bar{\theta}_{\text{base}}, \tilde{x}_i, u_i))_{i=1}^N$,

$$\begin{aligned} \cos \angle(v, \text{range } \Phi) &= \frac{\|\Phi \Phi^+ v\|}{\|v\|}, \\ \cos \angle(v, F_{\text{base}}) &= \frac{|\langle F_{\text{base}}, v \rangle|}{\|F_{\text{base}}\| \|v\|}. \end{aligned} \quad (33)$$

Applied to Δ^* , the first measure is zero exactly when the recovery condition holds. It needs θ_{base}^* , so it is available in simulation only. Applied to the corrected learned write on held-out tuples, the first measure shows how far the guarantee extends beyond the reference set. The second measures the overlap with the baseline output that the construction leaves unprotected. Section VI tests whether the split is start-independent and, only where the condition holds, whether the true parameters are recovered. **[TODO: Should the proposed orthogonal multisine design be a future-work result, or must it be run before claiming a constructive remedy?] [Moved out of this section: measurement details of the pointwise no-go result and the Coulomb inner-product argument (now bullets in Section VII, Limitations); rank, condition number and column correlations of J (Results, R4); arm comparison (Section V, Aug-U and Aug-OBC only). Dropped: $\rho^* = 0.205$ and the 4.96% floor (earlier absorber-only dataset, not the thesis truth).]**

V. EXPERIMENTAL SETUP

A. Simulation benchmark and excitation

- Generate closed-loop records from the physics baseline plus a truth-only flexible absorber, retaining exact states solely for diagnostics

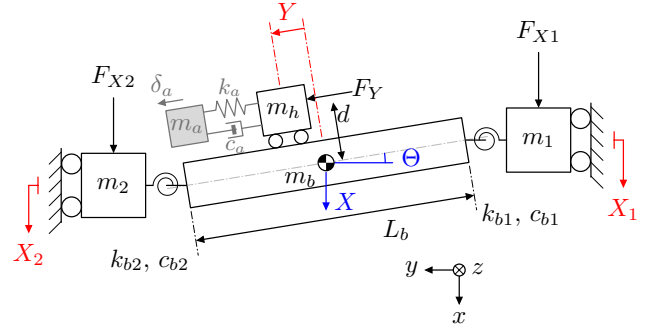


Fig. 5: Simulated gantry with a payload-mounted absorber. The light-gray elements (m_a, k_a, c_a, δ_a) are absent from the baseline.

- Use multisine excitation that covers the absorber pole and anti-resonance while respecting position, velocity, yaw, and force limits
- Assign independent random multisine phases to training, validation, and test records so equal spectra do not duplicate trajectories
- Include multisine-free cycloidal motion profiles matching the measured machine motion to test the operational distribution shift
- Use only the corrected all-channel Coulomb dataset matched to the frictionless excitation; the earlier mismatched-amplitude merge is superseded
- The set-valued friction law requires a velocity threshold below which a rail is treated as stuck. This is set to a physical breakaway velocity, $v_{\text{brk}} = 2.25 \cdot 10^{-3}$ m/s, taken from Lee et al. [23], Table 5, identified on a THK SSR20XW linear-motion guide. Reported Stribeck velocities for lubricated guideways span 10^{-5} to 10^{-2} m/s [23, Sec. 5.2], so the chosen value sits mid-range **[TODO: Conflicts with D-204, where the stick band is $V_{\text{EPS}} = (cc1+cc2)/m_{\text{total}} \cdot ts$ labelled THEORY: leine1998, about $3.3 \cdot 10^{-5}$ m/s before the 3 to 10x margin. Confirm which threshold the reported runs use, and whether the second citation (Stribeck range) is the same paper.]**
- State that output noise is currently disabled; add it only if a level and measurement location can be estimated from Jasper, ILC data, or stationary records
- State the sample rates, record lengths, number of windows, and total independent motion time rather than quoting overlapping-window count as data volume
- Decision (RK4 value, moved here from Section II): one reproducibility sentence giving the RK4 step, equal to the sampling period after downsampling the 20 kHz records to 4 kHz, with one substep (entry script: `fs_new=4000, up_sample=1`). Confirm against the `config.json` of each reported run, since earlier checkpoints may have used another `up_sample`
- Figure (reuse, no separate asset): let the rebuilt training-objective time strip also identify independent phases, short

training windows, encoder history, and the long evaluation horizon

[TODO: Is the hidden linear absorber sufficient simulation complexity, or is a truth-only cubic spring required to demonstrate nonlinear augmentation?]

B. Real Telica data and split

- Confirm with Jasper and Dragan which reconstructed ETEL setpoint-generator or ILC-reference motions are operationally representative, including minimum/maximum move, long stroke, acceleration, and dwell
- Use the supervisor-defined split of 11 training, two validation, and two test operating points at 20 kHz
- Convert rail positions and currents to logical metres and Newtons; form velocity states by central differences because direct velocity is not logged
- Keep all iterations at one operating point in the same split to prevent identical reference trajectories leaking across sets
- Describe the X and Y operating-point grid and show whether training covers the interpolation region occupied by both test points
- Use iteration 0 for controller checks and later ILC iterations for distinct feedforward conditions without treating them as independent trajectories
- Mark real-data training and held-out evaluation as planned until a final reproducible run and force-calibration convention exist
- Figure (new): 2D operating-point grid with train, validation, and test locations, operating-range boundary, and trajectory classes; add a 3D or pairwise state-input coverage view only if it reveals coverage hidden by the 2D map

[TODO: Freeze the approved operational trajectory set with Jasper and Dragan; should direct drive velocity be requested, and does a third coverage coordinate add information beyond the 2D grid?]

C. Training protocol, comparators, and metrics

- Match records, neural seeds, windows, optimiser budget, and checkpoint criterion across baseline-only, unconstrained, tangent-OBC, affine-OBC, and black-box arms; treat multisine phase seeds separately
- Use physical RMS as the primary candidate metric for tracking and a fixed 200–400 ms post-move settling interval; use NRMS, BFR, or spectra only for a distinct supporting question
- Report physical combination error, projection overlap, added-state ablations, and encoder-init sensitivity alongside predictive metrics
- Values referenced from Section III (moved 2026-10-06), each with an honest basis (derived, inherited, tuned, compute-limited): encoder history n , window length n_f and stride, number of additional states $n_{\bar{x}}$, network width and depth, model rate

- Free-run validation (referenced from Section III-C): whole records, encoder at the record start and $x^c = 0$ once, so model error accumulates over the record unlike in a training window; state the record-length-weighted RMS used for checkpoint selection
- Decision (moved from Section II-C): joint-estimation arms start from the nominal φ of the reference implementation detuned by 10% per combination (`combo_init_detune`, HEURISTIC, D-191), with the nominal-parameter prior disabled (D-193); state it as a run setting with its reason
- A parameter prior bounds the drift of the combinations but does not remove the negation flat direction of Section IV; it is off in every arm (D-193)
- Practical identifiability (referenced from Section II-C): report the rank of the stacked one-step transition Jacobian with respect to φ on the training data (rank 10, nullity 4 in θ per the reduced-block docstring), from a saved artefact rather than the docstring
- Value (moved from Section II-C): the margin $\rho = 0.99$ in (20), a HEURISTIC margin inside the admissible region (D-229)
- Metric (moved from Section II-C): relative combination error in θ_{base} , with the normalisation of the small signed m_{Δ} stated explicitly
- State per arm whether raw or reduced coordinates are trained; exact-model parameter recovery is an internal implementation check, not a result (moved from Section II-C)
- Give the black-box model validation-selected capacity and horizon so it is a capability comparison rather than an artificially weakened baseline
- Evaluate held-out position/phase interpolation, ASMPt-relevant multisine-free motion, controller transfer, and operating-range extrapolation as separate rungs
- Predeclare expectation, acceptance criterion, verdict, and mechanism for every result rather than presenting isolated scores

[TODO: Decide the primary metric, credible controller-transfer arm, and feasible neural-seed count; do not pool initialization, noise, phase, and operating-point uncertainty.]

D. Reproducibility and implementation checks

- Verify controller reconstruction, coordinate transforms, normalisation, RK4 transition, and LFR equivalence before training
- Use the same controller, data conversion, and rollout entry point for training, validation, and checkpoint selection
- Controller rate (moved from Section III-C, 2026-10-06): the records are generated with the controller at 20 kHz, the training loop re-discretises it at the 4 kHz model rate (D-141); the residual form of Section III-C equals the shared-reference loop only for a compatible controller, signals and initial state
- Compatibility check for that assumption: run the re-discretised controller on the recorded $r - y$ and compare

its feedback force with the recorded one; a rollout with $\hat{y} = y$ gives $\hat{u} = u$ by construction and checks nothing

- Record configuration, software precision, seed, dataset identifier, and checkpoint provenance for every reported arm
- Keep oracle states and truth parameters out of the trainable pipeline and label every diagnostic that uses them
- Figure (existing, appendix candidate): use `controller-chain-v1` only as an audit trail for controller reconstruction and discretisation, not as a main method figure

[**TODO:** Which implementation checks belong in the main paper, and which should move to the appendix or repository documentation?]

VI. RESULTS

A. Baseline and closed-loop formulation

- Expectation and criterion: the LPV-LFR and original physics equations agree numerically across the payload range before model-error comparisons are meaningful
- Verdict and reason: report inertia invertibility, LFR/RK4 equivalence, and the remaining baseline error caused by the truth-only absorber
- Figure: baseline and oracle output errors versus time for representative payload positions; supports the claim that the omitted mode is systematic and position dependent
- Table: baseline free-run physical RMS for LPV and frozen-LTI models across held-out Y ; tests the research plan's Aspect 1 rather than assuming LPV superiority
- Expectation and criterion: direct and residual closed-loop forms match at numerical precision when controller assumptions hold
- Evidence for Section III-C (moved 2026-10-06): on the thesis data, open-loop rollouts accumulate position error on the free axes X and Y , closed-loop rollouts do not; re-measure, do not reuse the older D-142 numbers
- Figure: recorded input, controller correction, and model error over one rollout; supports the interpretation that feedback can suppress visible output error while the open-loop model remains wrong

[**TODO:** The LPV-versus-frozen-LTI comparison is required by the plan but may not yet have a final matched result; run it or narrow Aspect 1 explicitly.]

B. Predictive benefit and additional-state contribution

- Expectation and criterion: augmentation must reduce held-out long-horizon error, not only the short training loss, relative to the same encoder-initialised baseline
- Verdict and reason: report the best validated improvement and the gap between window fit and free-run accuracy caused by integrated velocity bias
- Figure: validation RMS against optimiser updates for baseline and augmented arms, with window and 12 s metrics; supports or rejects objective alignment

- Figure: error spectra before and after augmentation with absorber pole and anti-resonance marked; tests whether the learned correction targets the missing band
- Figure: clamp each additional state and the full added-state vector to zero during rollout; quantifies what the trained latent dynamics add or harm
- Table: standard model, no-added-state ablation, zero-initialised latent path, and longer encoder history; connects structure changes to predictive benefit
- State the observed weak or harmful added-state contribution as a result if the final run does not reverse it, rather than inferring usefulness from state magnitude

[**TODO:** Are state Gramians useful for explaining each added state's contribution, or would intervention-based ablation be more direct and less linearisation-dependent?]

C. Parameter interpretability and orthogonality

- Expectation and criterion: OBC should reduce physical combination error without materially worsening held-out predictive fit relative to unconstrained joint estimation
- Verdict and reason: the completed arms reach 7.20% no projection, 7.69% tangent OBC, and 8.47% unfinished affine OBC from a common 9.49% start, so aggregate recovery does not establish OBC superiority
- Figure: validation fit versus combination error over training for all three arms; supports the claim that near-equal fit permits materially different physical parameters
- Figure: per-parameter error at matched budget; shows which combinations projection protects and which near-flat directions redistribute error
- Table: basis rank, condition, overlap, final and minimum parameter error, and validation RMS; links algorithm health to the recovery verdict
- Figure: ρ^* and predicted versus observed parameter bias by singular direction; supports the conditional result that true recovery is limited by non-orthogonal missing physics
- Explain why the 4.96% predicted floor accounts for part of the residual squared error but does not match the observed per-parameter pattern or prove convergence

[**TODO:** Should the incomplete affine arm be retained, rerun to a matched budget, or reported only as exploratory evidence?]

D. Generalisation and black-box comparison

- Expectation and criterion: improvement must progress from held-out position/phase interpolation to ASMPt-relevant motion, controller transfer, and quantified operating-range extrapolation without retuning
- Verdict and reason: separate completed simulation evidence from pending Telica results and explain any failure by excitation coverage, initialisation, or objective mismatch
- Figure: tracking overview plus a dedicated 200–400 ms settling zoom for baseline, augmented model, and black box on an approved operational trajectory

- Table: primary RMS, fixed-window settling RMS, parameter availability, and training data for baseline, augmented, and black-box models on every completed validation rung
- Figure or table: train with one controller and evaluate the frozen models with a credible changed controller; pair this with a plant-domain check for feedback masking
- Figure: error versus multisine amplitude, stroke, acceleration, scheduling position, or state-input distance from the training set; quantify extrapolation rather than calling all held-out data extrapolation
- Report the black box's early improvement and later free-run degradation as a negative comparator result if no stable matched run supersedes it
- Aggregate the final learned-model comparison across neural initialization seeds and show a representative, not best, time trace; report phase realizations separately

[TODO: Freeze which validation rungs are complete enough for a final claim; omit rather than speculate about unfinished controller, noise, or hardware experiments.]

VII. DISCUSSION

A. Answers to the four research aspects

- Aspect 1: state whether explicit Y scheduling earns measurable predictive value over frozen LTI baselines
- Aspect 2: distinguish the expressive value of dynamic augmentation from the observed trainability of its added states
- Aspect 3: conclude that OBC enforces a unique local split but true parameter recovery is conditional on the real discrepancy's orthogonality and optimiser convergence
- Aspect 4: compare simulation interpolation, motion-profile transfer, amplitude extrapolation, real-data evidence, and the black-box result without merging them into one score
- Treat controller transfer as the industrial digital-twin use case and as a closed-loop distribution shift distinct from operating-range extrapolation
- Interpretability matters when physical parameters support controller redesign or health monitoring, but its practical value must be tied to a named downstream decision
- Figure plan: introduce no new figure; interpret the architecture, OBC, and result figures already presented instead of adding a summary graphic

[TODO: Which result is the primary answer if predictive accuracy and parameter recovery favour different arms?]

B. Limitations and lessons

- Closed-loop feedback improves numerical trainability but can hide plant mismatch, suppress orthogonality overlap, and reward a model that is poor in open loop
- Short windows underweight velocity DC bias on free integrators; varying encoder and rollout length is therefore a structural sensitivity test, not routine tuning
- One strengthened absorber, currently noiseless simulation, a finite neural-seed campaign, and truth-only dry friction bound claims about nonlinear, noisy, and dissipative missing dynamics

- State the final seed count and corrected-friction evidence here; do not retain development-era caveats after the final campaign changes them
- Central-difference velocities, force calibration, and the train-validation-test coverage grid bound the credibility of any real-data conclusion
- The research plan expected validated real-machine generalisation; any missing Telica result is a departure to disclose, not replace with design documentation
- Augmentation versus residual-FRF fitting remains a trade between trajectory-domain nonlinear prediction and a simpler frequency-domain account of missing dynamics
- Pointwise orthogonality is impossible for any nonzero missing force on this gantry: the parameter sensitivity of the three velocity rows has rank 3 at every reference sample (worst $\sigma_3/\sigma_1 = 5.7 \cdot 10^{-4}$, earlier absorber dataset), so $J^\top \Delta^* = 0$ can only arise from cancellation over the record (Section IV-C)
- Coulomb friction in the truth violates the recovery condition by construction: the c_{g1} sensitivity is proportional to $\dot{X}_1$ and the friction is $-c_{c1} \text{sign}(\dot{X}_1)$, so their inner product is $-c_{c1} \sum |\dot{X}_1| \neq 0$ for any moving record (D-204); on T_AF, OBC gives a unique split, not unbiased combinations

[TODO: Should a cubic-spring benchmark, noise study, and Monte Carlo analysis be completed, or should the thesis explicitly narrow its claims to the measured linear absorber case?]

C. Implications and future work

- Design excitation so $J^\top \Delta^* = 0$ when true parameter recovery, rather than only a unique decomposition, is required
- Separate learning rates for physical and neural parameters and compare convergence against the predicted nonzero OBC bias floor
- Replace the short-horizon objective or couple windows so integrated velocity bias is visible during training
- Repair and rerun the Coulomb benchmark before claiming that the method learns a legitimate nonzero-mean residual
- Extend evaluation beyond the multisine range and use machine-relevant trajectories to delimit safe interpolation and extrapolation

[TODO: Which future-work item is a prerequisite for the thesis claim and which genuinely belongs after graduation?]

VIII. CONCLUSION

A. Conclusions

- Answer whether LPV-LFR dynamic augmentation improves dual-gantry settling prediction over the physics baseline and black-box comparator
- State whether the learned additional states contributed measurable missing dynamics under the final validated configuration
- Conclude that state-level OBC prevents local tangent overlap by construction, while true parameters are recovered

only when the physical discrepancy is orthogonal on the chosen excitation

- Give the final quantitative trade between prediction error and physical combination error on held-out data
- State whether the evidence covers relevant ASMP motion, settling, controller transfer, or only a narrower simulation validation ladder
- Close with the practical recommendation for when ASMP should use the augmented model and when the baseline or a simpler residual model is safer
- Figure plan: no conclusion figure; reserve the space for the final quantitative answer and limitations

[TODO: Rewrite every conclusion bullet after freezing the final result table; no planned or diagnostic result may appear as achieved.]

B. Discussion

- Why all states routing.
- why not use the encoder for the orthogonality each epoch instead of finite difference
- why not use the positions since we measure them $-i$ train nn and encoder in combination? not full answer
- black box not fair comparison, no init (BLA) no controller Thesis

Make a new folder for figures and code of the figures for the paper, and the final dataset that will be used.

* laten zien met parameters samenvoegen huidige methode niet correct ik moet iets van jacobian of iets laten zien * ik heb nog steeds niet netjes uitgewerkt voor LPV-LFR met hoe ik het heb omgeschreven die n m ik kan niet zo goed zeggen hoe het heet * verbeterde model met additional states laten zien als ik ze weg haal wat ze toevoegen = op nul zetten. ik weet niet of andere phase multisine validation en test genoeg laat zien hoe goed het werkt. * Duidelijk verantwoorden controller. vooral met integrator problem van de plant en dus X en Y geen stijfheid $K = 0$ daar. kleine integratie error propegeert gwn door, dit zei maarten ook denk dat dit de error is. * velocity central difference, why not get direct? * Zijn Gramians nuttig om per state te laten zien? * Multisine phase uitleggen kunnen we ook bronnen voor gebruiken. dat verschillende train validation en test verschillende multisine phase hebben. * Met alleen simulatie is huidige simulatie niet genoeg, cubic spring toevoegen?? * benefit van mijn closed-loop training method is ook dat ik geen encoder / states er voor hoeft te initializen. kessels wel referencen. figuur gebruiken. * Jacobian additional states * 2d / 3d grid wat jan zei van de data space of ik alles vang van het gebied van telica. * Montecarlo simulatie zoals jan? * For parameter recovery I had trajectories similar to the paper, currently not used for joint estimation? * How well do we extrapolate outside multisine range? what does jan do?

Verdedigen

* waarom geen basis functions * waarom niet zoals kessels: vooral andere literatuur kunnen zeggen waarom niet dat gebruikt, moet dat? * wat is het voordeel van interpreteerbaarheid?—i controller maken, monitor health? * waarom niet frf of mass lijn fitten voor de missende dynamica

* controller surpresses / hides issues such as orthogonality overlap. Byproduct of closed-loop. * Hoe verdedig ik augmentation over residual frf fit?? * Hoeveelheid data kunnen verdedigen * Huidige nf window and encoder window, should i run for different lengths? * Orthogonality definition and that we orthogonalize additional states with respect to the baseline := 0. additional states propagate, not included in our definition. * Kijken decisions.md * do i need to justify controller can i say what i got from my supervisor, how to approach these kind of questions?

APPENDIX A

SUPPORTING DERIVATIONS AND REPRODUCIBILITY

A. Explicit LPV-LFR realization

In (5), M_0 is (3) evaluated at $Y = 0$, and

$$M_1 = \begin{bmatrix} 0 & -m_h & 0 \\ -m_h & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \quad M_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & m_h & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

With a_1 and a_2 from (7) treated as independent signals, solving $M_0\ddot{q} + M_1a_1 + M_2a_2 = u - C\dot{q} - Kq$ for $\ddot{q}$ with $R = M_0^{-1}$ gives $\ddot{q} = R(u - Kq - C\dot{q} - M_1a_1 - M_2a_2)$. With $\tilde{x} = \text{col}(\dot{q}, \ddot{q})$, $z = \text{col}(\ddot{q}, a_1)$, and $w = \text{col}(a_1, a_2)$, the LTI part G of (10), partitioned as in [13, Eq. (6)], is

$$\begin{bmatrix} \tilde{x} \\ z \\ q \end{bmatrix} = \underbrace{\begin{bmatrix} A_x & B_w & B_u \\ C_z & D_{zw} & D_{zu} \\ C_y & D_{yw} & D_{yu} \end{bmatrix}}_G \begin{bmatrix} \tilde{x} \\ w \\ u \end{bmatrix}. \quad (34)$$

with blocks

$$\begin{aligned} A_x &= \begin{bmatrix} 0 & I_3 \\ -RK & -RC \end{bmatrix}, & B_w &= \begin{bmatrix} 0 & 0 \\ -RM_1 & -RM_2 \end{bmatrix}, & B_u &= \begin{bmatrix} 0 \\ R \end{bmatrix}, \\ C_z &= \begin{bmatrix} -RK & -RC \\ 0 & 0 \end{bmatrix}, & D_{zw} &= \begin{bmatrix} -RM_1 & -RM_2 \\ I_3 & 0 \end{bmatrix}, & D_{zu} &= \begin{bmatrix} R \\ 0 \end{bmatrix}, \\ C_y &= [I_3 \ 0], & D_{yw} &= 0, & D_{yu} &= 0. \end{aligned} \quad (35)$$

Here $\tilde{x}, z, w \in \mathbb{R}^6$ and $u, q \in \mathbb{R}^3$, so $G \in \mathbb{R}^{15 \times 15}$. The actuator frame enters only through $u = Pu_{\text{act}}$ and $q_{\text{act}} = P^\top q$ in (1).

Substituting $w = \Delta(Y)z$ into the z equation of (34) gives the algebraic loop equation $(I_6 - D_{zw}\Delta(Y))z = C_zx + D_{zu}u$. Written out,

$$\begin{bmatrix} I_3 + YRM_1 & YRM_2 \\ -YI_3 & I_3 \end{bmatrix} \begin{bmatrix} \ddot{q} \\ a_1 \end{bmatrix} = \begin{bmatrix} R(u - Kq - C\dot{q}) \\ 0 \end{bmatrix}.$$

The second block row gives $a_1 = Y\ddot{q}$, and the first then gives $M(Y)\ddot{q} = u - C\dot{q} - Kq$, that is (2), after multiplication by M_0 . The condition of [13, Def. 1], that $I_6 - D_{zw}\Delta(Y)$ is nonsingular, is therefore the condition that $M(Y)$ is nonsingular: the Schur complement of the lower identity block gives

$$\begin{aligned} \det(I_6 - D_{zw}\Delta(Y)) &= \det(I_3 + YRM_1 + Y^2RM_2) \\ &= \det M(Y) / \det M_0. \end{aligned}$$

Since $M(Y)$ is nonsingular, the loop is solved in closed form, $\ddot{q} = M(Y)^{-1}(u - C\dot{q} - Kq)$ with

$$M(Y)^{-1} = \frac{N_0 + YN_1 + Y^2N_2}{d(Y)}, \quad d(Y) = \det M(Y),$$

TABLE I: Nominal baseline parameters.

Parameter	Value	Parameter	Value
m_1	10.2 kg	m_2	10.7 kg
m_b	22.8 kg	m_h	10.1 kg
J_b	1.00 kg m ²	J_h	0.05 kg m ²
c_{g1}	14.5 N s/m	c_{g2}	20.3 N s/m
c_y	10.0 N s/m	d	0.10 m
c_{b1}	9.0 N m s/rad	c_{b2}	9.0 N m s/rad
k_{b1}	1987.5 N m/rad	k_{b2}	1987.5 N m/rad
L_b	0.725 m		

where $N_0 + YN_1 + Y^2N_2 = \text{adj } M(Y)$ and $d(Y)$ are both quadratic in Y . The implementation evaluates both polynomials with Horner's scheme. **[TODO: Missing: print N_0, N_1, N_2 and $d(Y)$ explicitly? Candidate: no, appendix budget; the quadratic degree was checked symbolically (sympy, 2026-10-06).]**

[TODO: Remarks cut from Section II-B; keep here or drop?]

- Y^2 is realized by repeating Y , not as a second scheduling variable $p_2 = Y^2$, which would admit pairs with $p_2 \neq Y^2$; extra scheduling variables cause overbounding in controller synthesis [13]
- The scheduling map is a known selection of the state ($Y = q_3$), not a learned map as in the self-scheduled models of [13]
- The exact condition $\det M(Y) \neq 0$ replaces the sufficient condition of [13, Thm. 6], which needs a scheduling set in the unit ball and $\rho(D_{zw}) < 1$
- Drenth's Def. 1 is stated in discrete time but constrains only the algebraic loop equation, so it applies unchanged in continuous time

B. Nominal physical parameters

Table I lists the nominal values used by the baseline. They are taken from the ETEL gantry reported by García-Herreros et al. [1, Table I]. Here θ collects the fourteen table entries other than the fixed length L_b .

C. Positive definiteness of the inertia matrix

With $a = L_b m_\Delta / 2$, elimination of the m_h block and completion of the square give

$$\begin{aligned} \frac{\det M(Y)}{m_h} &= (m_\Sigma + m_h)(J_{\text{eff}} + m_h Y^2) - (a - m_h Y)^2 \\ &= m_\Sigma m_h \left(Y + \frac{a}{m_\Sigma} \right)^2 \\ &\quad + (m_\Sigma + m_h) \left(J_{\text{eff}} - \frac{a^2}{m_\Sigma} \right). \end{aligned}$$

The 2×2 Schur complement has positive leading entry $m_\Sigma + m_h$. Sylvester's criterion therefore reduces positive definiteness to positivity of the completed-square expression for every Y . Hence, for positive m_h, m_Σ , and J_{eff} , $M(Y) \succ 0$ for all $Y \in \mathbb{R}$ if and only if (12) holds. For the physical parameterization, the difference between the two sides is

$$\begin{aligned} m_\Sigma J_{\text{eff}} - \frac{L_b^2}{4} m_\Delta^2 &= m_\Sigma (J_b + J_h) \\ &\quad + \frac{L_b^2}{4} [4m_1 m_2 + m_b (m_1 + m_2)], \end{aligned}$$

which is strictly positive for positive component masses and yaw inertias.

In (20), let $b(\theta_{\text{base}}) = \rho(2/L_b)\sqrt{m_\Sigma J_{\text{eff}}}$. The offset and scale are

$$r_0 = \frac{m_{\Delta,0}}{b(\theta_{\text{base},0})}, \quad \eta_0 = \text{atanh}(r_0), \quad s = \frac{m_{\Delta,0}}{b(\theta_{\text{base},0})(1 - r_0^2)}.$$

Here η_0 makes $\xi_\Delta = 0$ reproduce the initial combination, and s gives $\partial m_\Delta / \partial \xi_\Delta|_0 = m_{\Delta,0}$, matching the local relative-change scaling of the logarithmic coordinates. The construction requires $m_{\Delta,0} \neq 0$; a start near zero makes s small and the coordinate numerically inert. The exact inverse used for reporting is

$$\xi_\Delta = \frac{\text{atanh}(m_\Delta / b(\theta_{\text{base}})) - \eta_0}{s}.$$

D. OBC and experiment details

- Derive the stacked projector, rank-deficient pseudoinverse case, epoch refresh, and additional-state zero extension
- Derive $J^+ \Delta^*$ and list the measured singular-direction contributions to the predicted parameter bias
- List dataset identifiers, split membership, hyperparameters, seed, and exact metric definitions for every reported arm
- Figure (existing, optional): place `controller-chain-v1` here only if the controller reconstruction cannot be audited adequately in prose and equations

[TODO: Can code and data tables be cited through a repository or DOI instead of spending paper space on full reproducibility metadata?]

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